

PVision AI: Photovoltaic Fault Image Dataset Audit

Complete Inventory of Image Sample Counts, Folder Origins, and Modality Breakdown for AI / ChatGPT Handover

TOTAL IMAGES 69,484 Across 2 Modalities	GASF MODALITY 35,000 256x256 RGB	I-V MODALITY 34,484 1343x808 RGB	FAULT CLASSES 7 Classes 1 Normal + 6 Faults	DATA INTEGRITY 100% 0 Corrupt, 0 Dups
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1. Primary Storage Locations & Folder Hierarchy

The project dataset resides physically on disk under the following root paths. Each modality is partitioned into seven distinct subfolders named strictly after the physical fault category.

Directory Role	Physical Origin Path on Disk	Image Modality
Raw Dataset Root	D:\SolarPredict\dataset\gasf_images	GASF (Gramian Angular Field)
Raw Dataset Root	D:\SolarPredict\dataset\iv_images	I-V Characteristic Curves
Project Data Root	D:\SolarPredict\PVision_AI\data\pv_fault\gasf_images	GASF (Mirrored / Linked)
Project Data Root	D:\SolarPredict\PVision_AI\data\pv_fault\iv_images	I-V Curves (Mirrored / Linked)

2. Detailed Folder Origin & Image Count Breakdown

Below is the exact audit of every individual folder origin, sample count, percentage share, native dimensions, and sequence validation status:

Modality	Class Name	Folder Origin (Relative to dataset/)	Image Count	Share	Resolution	Audit Status
GASF	crack	gasf_images/crack	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	global_aging	gasf_images/global_aging	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	hotspot	gasf_images/hotspot	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	normal	gasf_images/normal	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	partial_aging	gasf_images/partial_aging	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	shading	gasf_images/shading	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF	short_circuit	gasf_images/short_circuit	5,000	14.29%	256x256 RGB	Complete (1-5000)
GASF Subtotal	7 Classes	gasf_images/ (7 folders)	35,000	50.37%	256x256	100% Balanced
I-V Curves	crack	iv_images/crack	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	global_aging	iv_images/global_aging	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	hotspot	iv_images/hotspot	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	normal	iv_images/normal	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	partial_aging	iv_images/partial_aging	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	shading	iv_images/shading	5,000	14.50%	1343x808 RGB	Complete (1-5000)
I-V Curves	short_circuit	iv_images/short_circuit	4,484	13.00%	1343x808 RGB	516 files missing (1-513, 597-599)
I-V Subtotal	7 Classes	iv_images/ (7 folders)	34,484	49.63%	1343x808	4,484 short_circuit
GRAND TOTAL	7 Unique Classes	14 Total Folder Origins	69,484	100.0%	RGB PNG	0 Corrupt, 0 Dups

3. ChatGPT Ready-to-Copy Specification Block

Copy and paste the formatted block below directly into ChatGPT to provide complete context of the image dataset:

Dataset Specification for PVision AI Fault Classification:

- Total Images: 69,484 PNG files across 14 directory origins.
- Modality 1 (GASF Images): 35,000 images, 256x256 RGB, 7 balanced classes (5,000 each: crack, global_aging, hotspot, normal, partial_aging, shading, short_circuit). Origin: D:\SolarPredict\dataset\gasf_images\
- Modality 2 (I-V Curves): 34,484 images, 1343x808 RGB, 7 classes (5,000 for 6 classes, 4,484 for short_circuit due to 516 missing index sequence numbers: 1-513, 597-599). Origin: D:\SolarPredict\dataset\iv_images\
 - Data Quality: 0 corrupted files, 0 duplicate images (MD5 verified), 100% sample retention.
 - Target Preprocessing: Resize to 224x224 RGB, ImageNet standardization, 70% Train (48,638), 15% Val (10,423), 15% Test (10,423) stratified split.
 - Objective: Train a Convolutional Neural Network (e.g. ResNet-18 or MobileNetV3) for 7-class PV fault diagnosis.